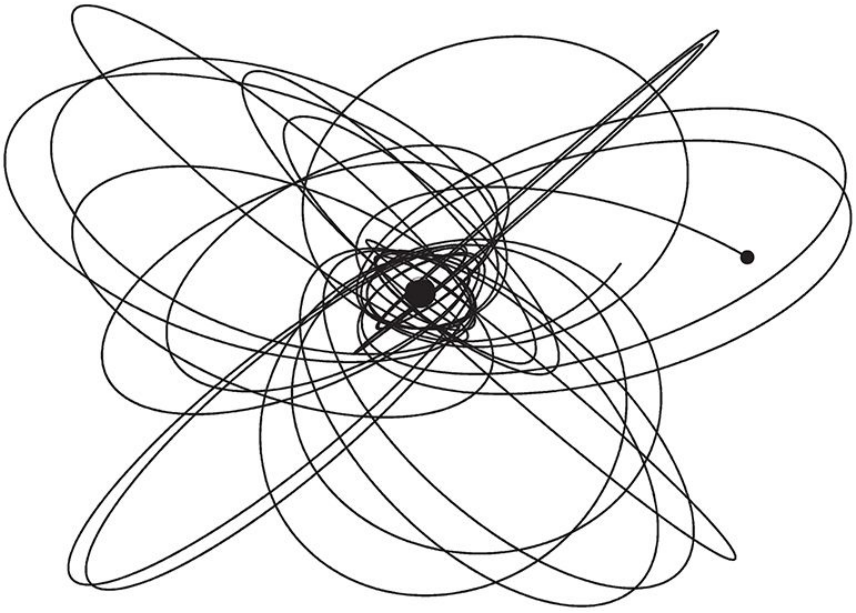


Fig. 10.1. Typical orbit of an object around a fast-spinning black hole like Gargantua. *[From a simulation by Steve Drasco.]*

With Gargantua and its dangerous environment under our belts, let's make a brief change of direction: to Earth and our solar system; to disaster on Earth and the extreme challenge of escaping disaster via interstellar travel.

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23 The angular momentum is the object's circumferential speed multiplied by its distance from Gargantua; and this angular momentum is important because it is constant along the object's orbit, even if the orbit is complicated.